Yiğit Efe Avcı — Statistics & Data Science

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Education

Istanbul Commerce University — Istanbul, Türkiye
B.Sc. in Statistics — 2023 – 2027
Anadolu University — Eskişehir, Türkiye
Associate Degree in Computer Programming — 2025 – 2027

Profile

Statistics undergraduate with a strong focus on data science, statistical modeling, data pipeline architecture, and empirical fieldwork. Possesses hands-on experience utilizing modern computation and machine learning libraries (Pandas, Polars, Scikit-learn) across Python, R, and SPSS workflows. Adeptly fuses data quality and survey methodology insights gathered from large-scale fieldwork with solid software engineering capabilities to design robust, end-to-end analytical solutions and automated retrieval tools.

Experience

Istanbul Chamber of Commerce — Istanbul, Türkiye
Long-Term Statistics Intern, İTOSAM Statistics Unit — 13 Oct 2026 – Present
Statistics Summer Intern, İTOSAM Statistics Unit — 10 Aug 2026 – 2 Oct 2026

- Actively contribute to the regular tracking and calculation of economic and commercial indicators, and assist in the development of index sets within the Strategic Research Center (İTOSAM).
- Perform time series analysis and seasonal adjustments on economic datasets to support the preparation of periodic statistical bulletins and reports.
- Contribute to data quality management, validation processes, and statistical methodology research to ensure data accuracy and reliability.

Ministry of Family and Social Services of Türkiye — Istanbul, Türkiye
Archive Officer, Beyoğlu Law Unit — 5 Jun 2026 – 7 Aug 2026

- Handled the archiving, registration, and tracking of legal case files and documents at the Istanbul Beyoğlu Law Unit.

Konda Research and Consultancy — Türkiye
Field Researcher — Jan 2025

- Managed direct on-field data collection workflows for nationwide socio-economic research initiatives and large-scale public sentiment surveys.
- Adhered strictly to rigorous sampling boundaries and methodological standards to ensure manipulation-free, high-fidelity data feeds for downstream statistical analysis pipelines.

Projects

Kurumzeka — AI-powered corporate search and data classification system — Python, Pandas, PostgreSQL, Redis

- Engineered a dynamic scoring and weighted filtering algorithm that programmatically evaluates unstructured search engine API nodes (Title, URL, Snippet) against institutional criteria.
- Implemented data engineering pipelines leveraging Pandas and Polars to execute comprehensive data cleaning, anomaly detection, and structured formatting optimized for LLM token boundaries.
- Integrated LLM-based autonomous AI Agents for real-time unstructured data categorization, utilizing Redis for asynchronous process queuing and cache layer optimization.

Machine Learning Analysis and Prediction of Public Transit Line Density in High-Demand Areas (TÜBİTAK 2209-A)
Python, Scikit-learn, Pandas, Open Data

- Developed predictive machine learning models to forecast commuter density across major transit nodes utilizing spatio-temporal datasets harvested from the Istanbul Metropolitan Municipality open data ecosystem.
- Processed spatial spatial distribution metrics to simulate data-driven route optimization alternatives and fleet allocation models to mitigate critical transit choke points.

Deckplane — Distributed Docker Management and Live Infrastructure Platform — Bun, Hono, TypeScript, WebSockets

- Architected a high-throughput control-plane / host-agent network topology powered by WebSocket and REST layers to monitor and manage distributed server nodes and microservices.

- Integrated automated Git-to-deployment infrastructure loops, granular Role-Based Access Control (RBAC) security frameworks, cryptographic audit logging, and automated OpenAPI 3.0 compilation.

Certifications

- **Building with the Claude API** — Anthropic, Jun 2026. Credential ID: q9kvcnmcziw7.
- **Introduction to Model Context Protocol (MCP)** — Anthropic, Jun 2026. Credential ID: pqvnmzrtup72.
- **Introduction to Agent Skills** — Anthropic, Jun 2026. Credential ID: 8jzyuoz3n3rw.
- **Claude Code in Action** — Anthropic, Jun 2026. Credential ID: wdv3rdszeb4c.
- **Machine Learning with Python** — IBM, Feb 2025. Credential ID: 1H9MA1TRY556.
- **AI Agent Development and MCP Fundamentals** — Cyber Clubs Union, Jan 2026. Credential ID: SKB-2026-733OF.
- **Zero-to-Advanced Python Programming** — BTK Akademi, Jan 2025. Credential ID: OKMhqXvw04.
- **Google Flutter Mobile Application** — BTK Akademi, Jan 2025. Credential ID: L8dcn9b8V6.

Skills

Data Science & Statistics: Pandas, Polars, NumPy, Scikit-learn, TensorFlow, PyTorch, SPSS, Statistical Modeling, Regression Analytics, Time-Series Forecasting, Data Visualization (Matplotlib, Seaborn)
Programming Languages: Python, R, Go, TypeScript, JavaScript, SQL, Dart
Backend & Web Technologies: Node.js, Bun, Next.js, Hono, REST API, WebSockets
Database & Infrastructure: PostgreSQL, Redis, MongoDB, NATS JetStream, Docker, Drizzle ORM, Git, Postman